

Rental Marketplace Platform

Project Description

The Rental Marketplace is a comprehensive peer-to-peer (P2P) and business-to-consumer (B2C) digital platform that allows users to list, discover, rent, and manage a wide variety of items. The system facilitates secure transactions, real-time communication, and reliable tracking of rental lifecycles. It is designed to maximize item utilization, provide individuals with passive income streams, and offer renters a cost-effective alternative to purchasing. The platform accommodates three distinct user roles: Lessors (Item Owners), Lessees (Renters), and Administrators.

Authentication Features (Not Counted as Functional Requirements)

- Users can register an account by providing personal details, verifying their identity, and selecting a primary user persona.
- Users can securely log in and log out using encrypted credentials.
- The platform supports automated password recovery via verified email or phone-based OTP tokens.
- The system provides multi-factor authentication (MFA) capabilities during the login phase to ensure account integrity.

Functional Requirements

1. The system supports three distinct account profiles—Lessee (Renter), Lessor (Owner), and Administrator—each with custom access privileges and interface views.
2. Users can build, update, and manage comprehensive profiles including personal bios, verification status, contact details, and location preferences.
3. Lessors can add new rental listings by inputting title, description, category, rental rate (hourly/daily/weekly), security deposit, and item rules.
4. Lessors can dynamically edit active listing attributes, adjust seasonal pricing tiers, and update availability schedules.
5. Lessors can archive or permanently delete listings from the public marketplace, provided there are no active or pending bookings for that item.
6. The system allows lessors to upload, organize, and set a primary preview image for high-resolution photos of the rental items.
7. Lessees can browse categorized rental listings through an intuitive UI

- featuring chronological and popularity sorting options.
8. The system features a robust search bar allowing users to query items using keywords, locations, and semantic tags.
 9. Lessees can apply multi-layered filters to search results, narrowing items down by category, price range, geo-location radius, and availability dates.
 10. Lessees can initiate rental bookings by specifying start/end periods and reviewing a breakdown of rental costs, deposits, and service fees.
 11. The system maintains a detailed Rental History log where users can view past, active, cancelled, and completed rental transactions.
 12. Users can save items to a personal Wishlist/Favorites archive to track availability changes or save listings for future rental consideration.
 13. The platform provides an integrated, real-time in-app chat system enabling lessors and lessees to discuss item specificities and coordination details safely.
 14. The system delivers automated notifications via in-app alerts and emails covering booking updates, payment status, and upcoming deadlines.
 15. Users can rate and submit qualitative reviews for both counter-parties and specific items upon the successful conclusion of a rental cycle.
 16. The system includes a reporting mechanism allowing users to flag fraudulent listings, damaged goods, or abusive behavior for admin intervention.
 17. The system tracks real-time payment status (Pending, Escrowed, Disbursed, Refunded) linked directly to third-party secure payment processing.
 18. The platform handles structured return confirmations, requiring both parties to verify item condition before the security deposit is settled.
 19. Administrators are equipped with user moderation tools to suspend accounts, verify identity documentation, and resolve user conflicts.
 20. Administrators possess absolute listing moderation capabilities to approve new merchants, delete policy-violating items, or feature specific listings.
 21. The system provides an analytical dashboard displaying personalized metrics such as item performance, total revenue, utilization rates, and operational activity.

Modules

Module 1: User & Item Management

- **Feature 1:** The system supports three distinct account profiles—Lessee (Renter), Lessor (Owner), and Administrator—each with custom access privileges and interface views.
- **Feature 2:** Users can build, update, and manage comprehensive profiles

including personal bios, verification status, contact details, and location preferences.

- **Feature 3:** Lessors can add new rental listings by inputting title, description, category, rental rate (hourly/daily/weekly), security deposit, and item rules.
- **Feature 4:** Lessors can dynamically edit active listing attributes, adjust seasonal pricing tiers, and update availability schedules.
- **Feature 5:** Lessors can archive or permanently delete listings from the public marketplace, provided there are no active or pending bookings for that item.
- **Feature 6:** The system allows lessors to upload, organize, and set a primary preview image for high-resolution photos of the rental items.

Module 2: Discovery, Search & Booking

- **Feature 7:** Lessees can browse categorized rental listings through an intuitive UI featuring chronological and popularity sorting options.
- **Feature 8:** The system features a robust search bar allowing users to query items using keywords, locations, and semantic tags.
- **Feature 9:** Lessees can apply multi-layered filters to search results, narrowing items down by category, price range, geo-location radius, and availability dates.
- **Feature 10:** Lessees can initiate rental bookings by specifying start/end periods and reviewing a breakdown of rental costs, deposits, and service fees.
- **Feature 11:** The system maintains a detailed Rental History log where users can view past, active, cancelled, and completed rental transactions.

Module 3: Communication, Social & Safety

- **Feature 12:** Users can save items to a personal Wishlist/Favorites archive to track availability changes or save listings for future rental consideration.
- **Feature 13:** The platform provides an integrated, real-time in-app chat system enabling lessors and lessees to discuss item specificities and coordination details safely.
- **Feature 14:** The system delivers automated notifications via in-app alerts and emails covering booking updates, payment status, and upcoming deadlines.
- **Feature 15:** Users can rate and submit qualitative reviews for both counter-parties and specific items upon the successful conclusion of a rental cycle.
- **Feature 16:** The system includes a reporting mechanism allowing users to flag fraudulent listings, damaged goods, or abusive behavior for admin intervention.

Module 4: Transaction Settlement, Analytics & Administration

- **Feature 17:** The system tracks real-time payment status (Pending, Escrowed, Disbursed, Refunded) linked directly to third-party secure payment processing.
- **Feature 18:** The platform handles structured return confirmations, requiring both parties to verify item condition before the security deposit is settled.
- **Feature 19:** Administrators are equipped with user moderation tools to suspend accounts, verify identity documentation, and resolve user conflicts.
- **Feature 20:** Administrators possess absolute listing moderation capabilities to approve new merchants, delete policy-violating items, or feature specific listings.
- **Feature 21:** The system provides an analytical dashboard displaying personalized metrics such as item performance, total revenue, utilization rates, and operational activity.